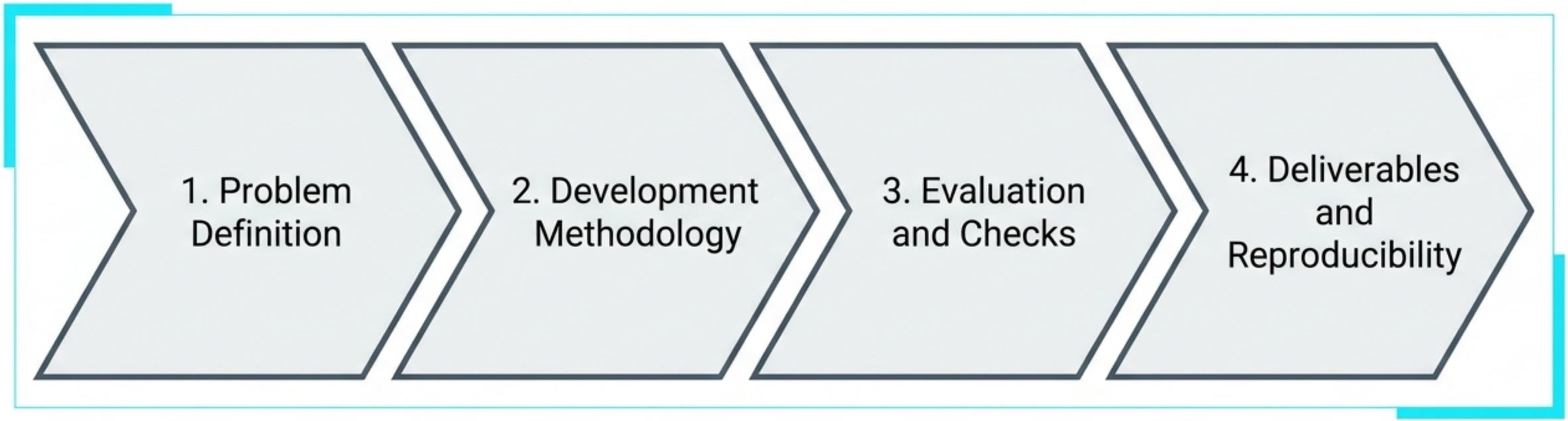
MODULE 16, LESSON 2

AI 101 for Mechanical Engineers:
A small project you can defend

Gemini Notebook

Lesson Outline



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Section 1: Problem and Data Definition

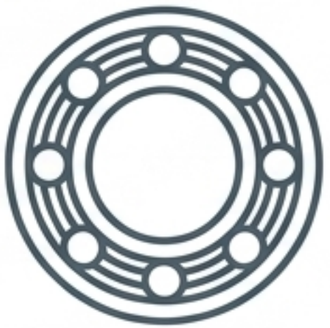
Purpose: Establish strict project boundaries, required inputs, target variables, and data provenance.

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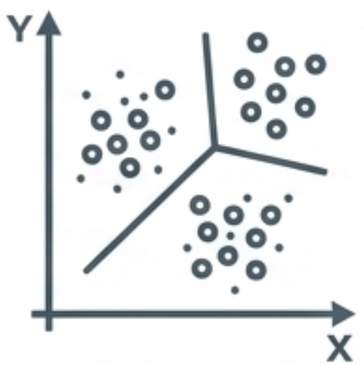
Permissible Project Tasks



Predict electrical pump power.



Classify supplied bearing scores.



Group operational machine profiles.



Critique AI-assisted engineering calculations.

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Required Stage 1 Evidence

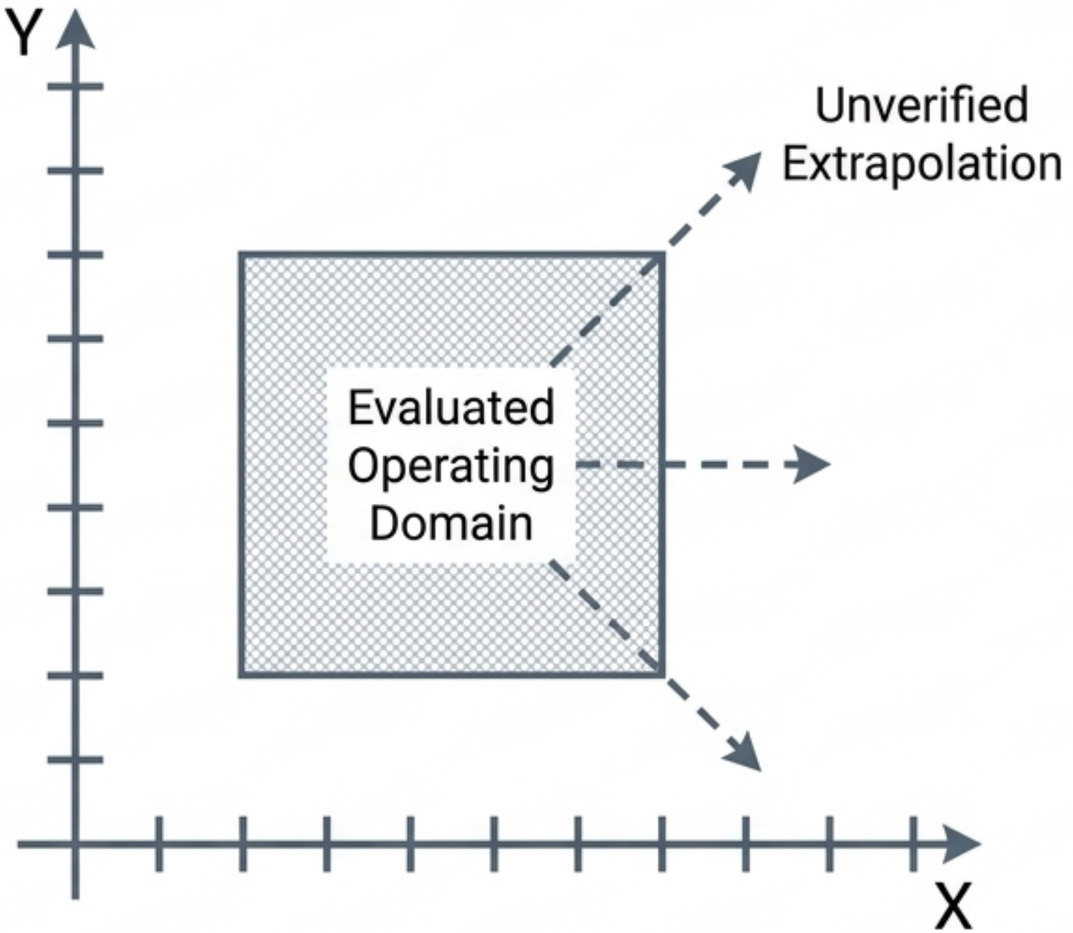
- Specify the engineering decision and users.
- Declare available input features explicitly.
- State the exact target variable.
- Document precise data provenance.



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Defining the Operating Domain

- Identifies conditions evaluated for suitability.
- Requires exact specification of machines.
- Requires defined mechanical load ranges.
- Precludes unverified extrapolation beyond limits.



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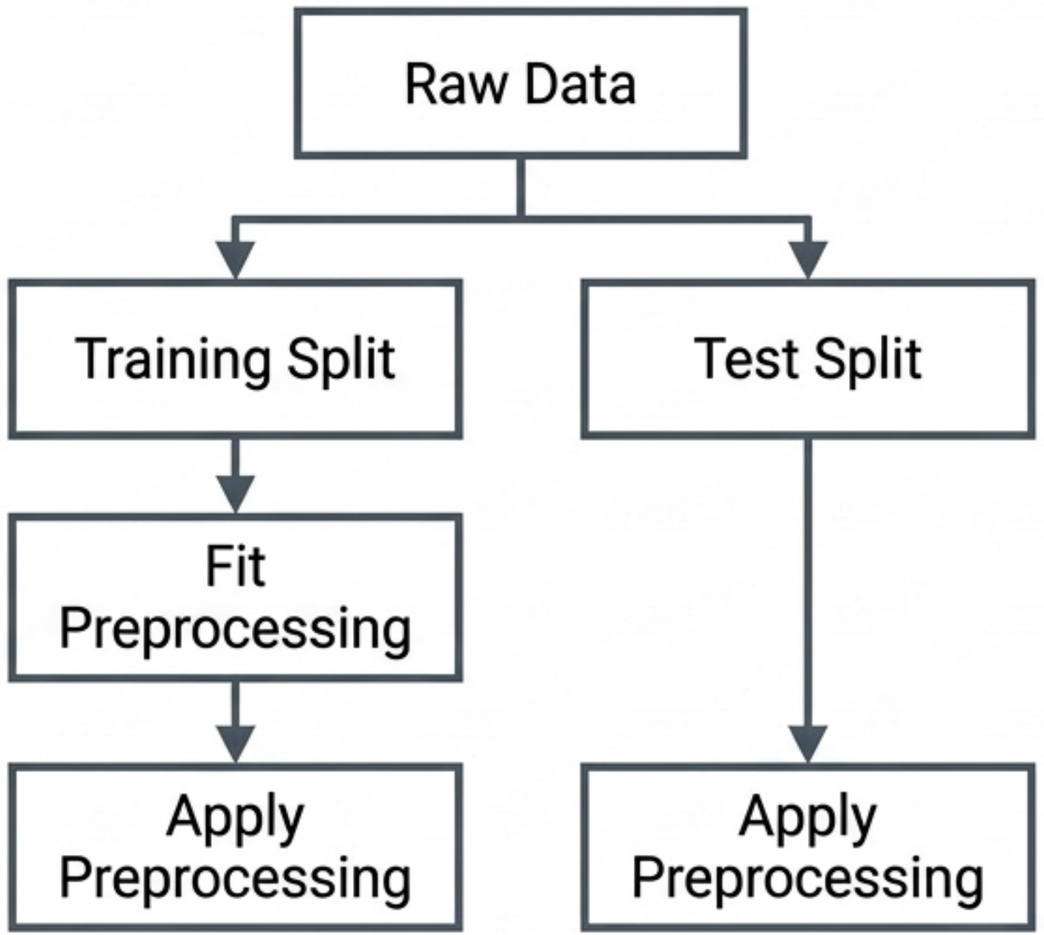
Section 2: Development Methodology

Purpose: Construct predictive baselines and execute rigorous, independent data splits.

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Evaluation Splits and Preprocessing

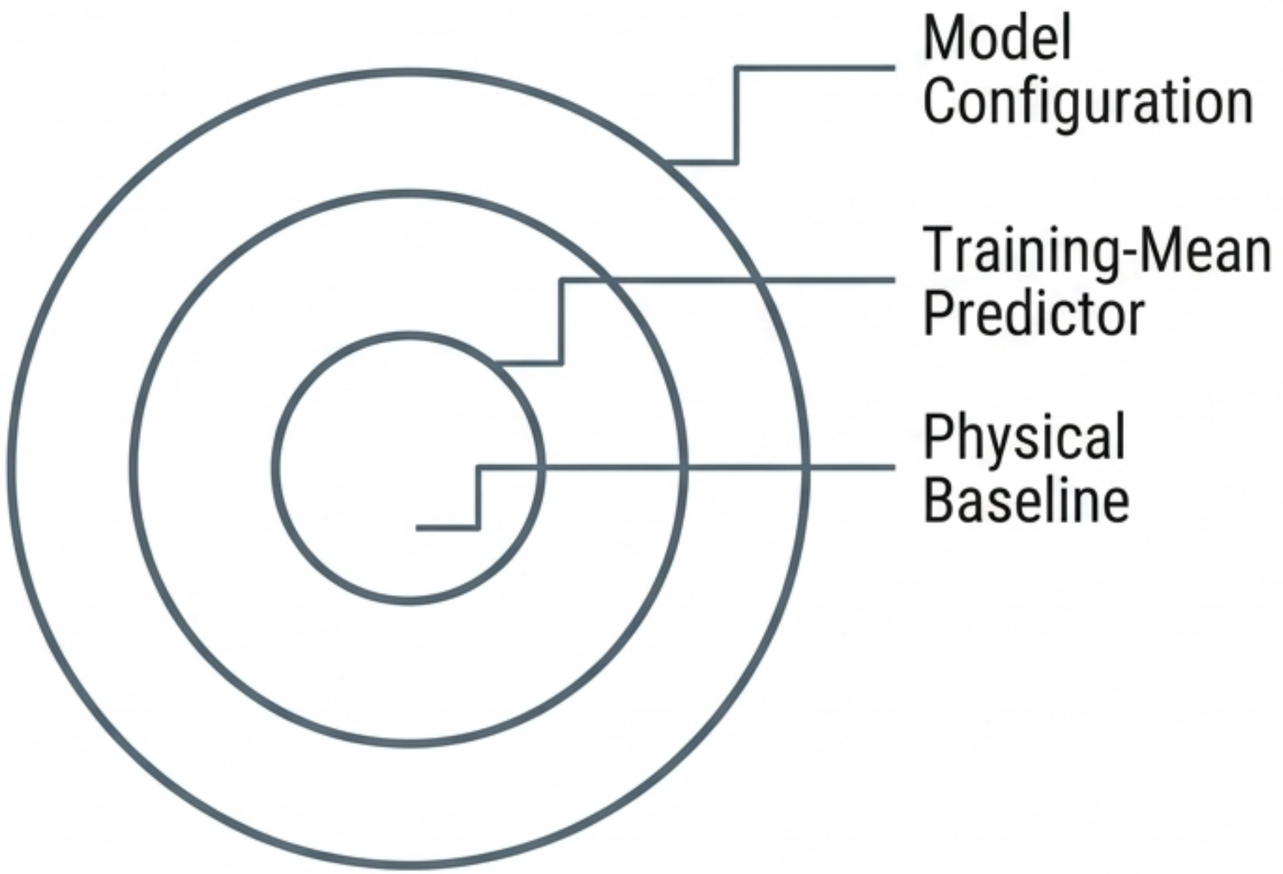
- Implement an independent evaluation split.
- Isolate test data before preprocessing.
- Fit preprocessing transformations on training data.
- Apply fitted transformations to test data.



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Required Comparative Baselines

- Compare models against training-mean predictors.
- Compare models against known physical baselines.
- Use validation subsets for configuration choices.



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Section 3: Evaluation and Checks

Purpose: Measure model performance against locked test data and independent physical realities.

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Final Error Reporting

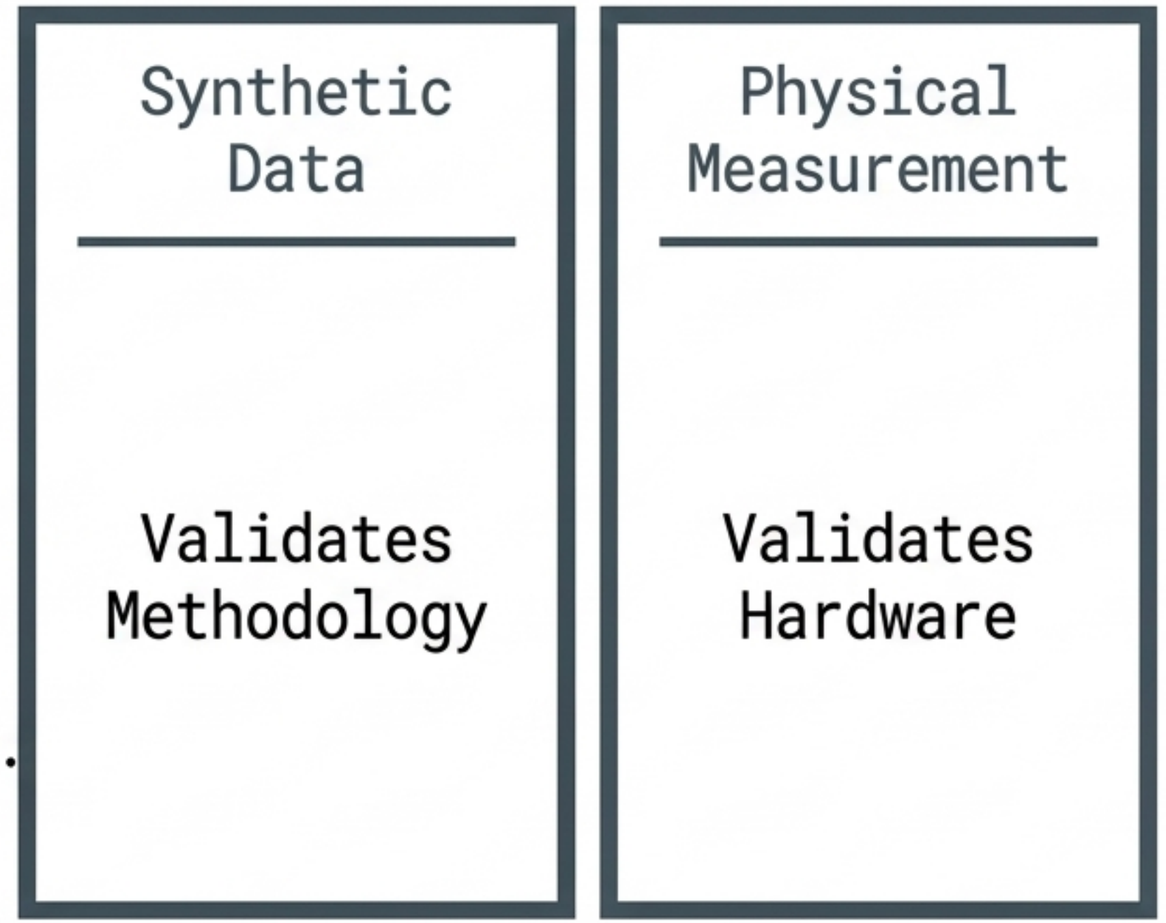
- Evaluate strictly against locked test sets.
- Report Mean Absolute Error (MAE).
- Report Root Mean Square Error (RMSE).
- Report all residuals with explicit SI units.

Metric	Calculation	SI Unit Requirement
MAE	Mean Absolute Error	Required (e.g., kW)
RMSE	Root Mean Square Error	Required (e.g., kW)
Residuals	Observed - Predicted	Required (e.g., kW)

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Hardware Versus Synthetic Validation

- Execute independent numerical or physical checks.
- Use synthetic data to illustrate methodologies.
- Note synthetic data precludes hardware validation.
- Document all identified physical limitations explicitly.



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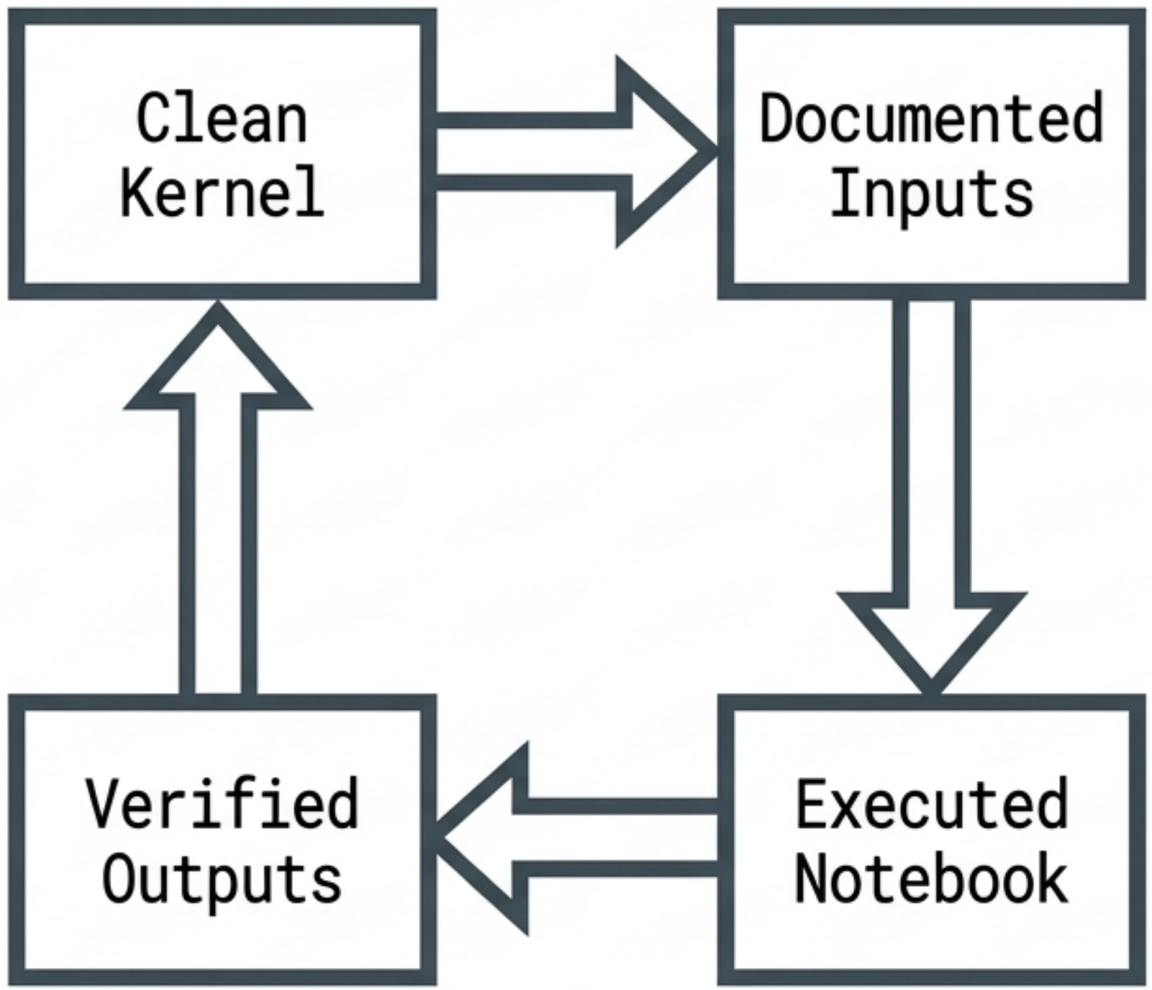
Section 4: Deliverables and Reproducibility

Purpose: Guarantee methodological transparency and code reproducibility for independent auditing.

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Ensuring Project Reproducibility

- Regenerate procedures using documented inputs.
- Ensure notebooks execute from clean kernels.
- Explain starter-code modifications in original words.
- Record permitted AI assistance explicitly.



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Key Points

Section 1: Define bounded tasks with explicit provenance and operational domains.

Section 2: Implement justified evaluation splits; fit preprocessing strictly on training data.

Section 3: Report locked-test errors and independent physical checks using SI units.

Section 4: Ensure exact reproducibility using runnable notebooks executed from clean kernels.

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